

1.2] Proportion, Variation & Partnership Topics

- 1) Proportion Concept**
- 2) Examples on Proportion**
- 3) Thought Process**
- 4) Theory Points in Proportion**
- 5) Variation**
- 6) Examples on Variation**
- 7) Partnership**
- 8) Homework Questions**
- 9) Homework Question Solution**

1] Proportion

$$\textcircled{1} \quad \frac{a}{b} = \frac{c}{d} \rightarrow \text{Proportion}$$

$$\therefore ad = bc$$

$$\textcircled{2} \quad a:b :: c:d \Leftrightarrow \frac{a}{b} = \frac{c}{d}$$

$$\text{Ex: } \frac{a}{b} = \frac{1}{2}$$

$$c:d = 10:20 = 1:2$$

$$\therefore \frac{a}{b} = \frac{c}{d}$$

③ Continued proportion

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e} \dots$$

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PROPERTY : $\frac{a}{b} = \frac{c}{d}$

① Componendo : $\frac{a+b}{b} = \frac{c+d}{d}$

② Dividendo : $\frac{a-b}{b} = \frac{c-d}{d}$

③ Componendo - Dividendo : $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

④ Invertendo : $\frac{b}{a} = \frac{d}{c}$

⑤ Alternendo : $\frac{a}{c} = \frac{b}{d}$

2] EXAMPLES ON PROPORTION:

Ex1: If $a:5 = b:7 = c:8$ then $\frac{a+b+c}{a} = ?$

Solⁿ: ① $\frac{a}{5} = \frac{b}{7} = \frac{c}{8} = k$

$$a = 5k, \quad b = 7k, \quad c = 8k$$

$$\textcircled{2} \quad \frac{a+b+c}{a} = \frac{5k + 7k + 8k}{5k}$$

$$= \frac{5+7+8}{5} = \frac{20}{5} = 4 //$$

Ex2: If $p:q = r:s = t:u = 2:3$ then $(mp+nr+ot):(mq+ns+ou) = ?$

Solⁿ: ① $\frac{p}{q} = \frac{r}{s} = \frac{t}{u} = \frac{2}{3}$

$$p = \frac{2}{3}q, \quad r = \frac{2}{3}s, \quad t = \frac{2}{3}u$$

$$\textcircled{2} \quad \frac{mp + nr + ot}{mq + ns + ou} = \frac{m\left(\frac{2}{3}q\right) + n\left(\frac{2}{3}s\right) + o\left(\frac{2}{3}u\right)}{mq + ns + ou}$$

$$= \frac{m\left(\frac{2}{3}q\right) + n\left(\frac{2}{3}s\right) + o\left(\frac{2}{3}u\right)}{mq + ns + ou}$$

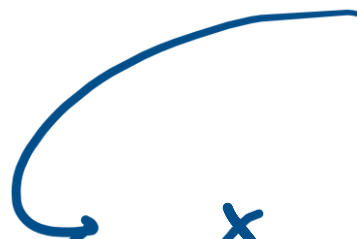
$$= \frac{2}{3} \frac{\cancel{mq + ns + ou}}{\cancel{mq + ns + ou}}$$

$$= \frac{2}{3} //$$

Ex3: If $x:y=y:z=z:u$ and $x:u = 64:27$. The value of $x:z$ is ?

Solⁿ : ① $x:y = y:z = z:u = k$

$$\frac{x}{u} = \frac{64}{27}$$


$$\frac{x}{y} \cdot \frac{y}{z} \cdot \frac{z}{u} = \frac{64}{27}$$

$$k \cdot k \cdot k = \frac{64}{27}$$

$$k^3 = \frac{64}{27} \Rightarrow k = \frac{4}{3}$$

$$\textcircled{2} \quad \frac{x}{z} = \frac{x}{y} \cdot \frac{y}{z} = k \cdot k = k^2 = \frac{4^2}{3^2} = \frac{16}{9} //$$

Ex4: A sum of Rs 427 is to be divided among A, B and C in such a way that 3 times A's share, 4 times B's share and 7 times C's share are all equal. Find the share of each

Solⁿ:

① $A + B + C = 427$

$$3A = 4B = 7C = K$$

$$\therefore A = \frac{K}{3}, \quad B = \frac{K}{4}, \quad C = \frac{K}{7}$$

$$\therefore A : B : C = \frac{K}{3} : \frac{K}{4} : \frac{K}{7}$$

$$\therefore A : B : C = \frac{K}{3} : \frac{K}{4} : \frac{K}{7}$$

$$= \frac{1}{3} : \frac{1}{4} : \frac{1}{7}$$

$$= \frac{84}{3} : \frac{84}{4} : \frac{84}{7}$$

$$A : B : C = 28 : 21 : 12$$

②

$$A + B + C = \text{total} = 427$$

$$A : B : C = 28 : 21 : 12$$

$$A = \frac{28}{28 + 21 + 12} \times 427 = 196$$

$$B = \frac{21}{61} \times 427 = 147$$

$$C = \frac{12}{61} \times 427 = 84$$

Ex5: $\frac{a-b}{a+b} = \frac{1}{5}$. Find $\frac{a^2-b^2}{a^2+b^2}$

Solⁿ: ① $\frac{a^2-b^2}{a^2+b^2} = \frac{(a^2-b^2)/b^2}{(a^2+b^2)/b^2}$

$$= \frac{a^2/b^2 - 1}{a^2/b^2 + 1}$$

(M1)
{concept}

(m1)

$$\frac{a-b}{a+b} = \frac{1}{5}$$

$$\rightarrow 5a - 5b = a + b$$

$$4a = 6b$$

$$a/b = 6/4 = 3/2$$

(m2)

$$\frac{a-b}{a+b} = \frac{1}{5}$$

$$\frac{(a-b) + (a+b)}{(a-b) - (a+b)}$$

$$= \frac{1+S}{1-S}$$

... { Compendo -
Dividendo }

$$\frac{(a-b) + (a+b)}{(a-b) - (a+b)} = \frac{1+s}{1-s}$$

$$\frac{2a}{-2b} = \frac{6}{-4}$$

$$\frac{a}{b} = \frac{6}{4} = \frac{3}{2} //$$

$$\text{Ans} = \frac{a^2/b^2 - 1}{a^2/b^2 + 1}$$

$$\frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2$$

$$= \frac{(a/b)^2 - 1}{(a/b)^2 + 1}$$

$$= \frac{(3/2)^2 - 1}{(3/2)^2 + 1} = \frac{(9 - 4)/4}{(9 + 4)/4} = \frac{5}{13}, //$$

Ex5: $\frac{a-b}{a+b} = \frac{1}{5}$. Find $\frac{a^2-b^2}{a^2+b^2}$

(mv)
{ substitution }

① Let $a=3, b=2$

$$\therefore \frac{a-b}{a+b} = \frac{1}{5} \quad \checkmark$$

②
$$\frac{a^2-b^2}{a^2+b^2} = \frac{(3)^2 - (2)^2}{(3)^2 + (2)^2}$$

$$= \frac{9-4}{9+4} = \frac{5}{13} //$$

3] Thought Process – {Learnings from Examples}

Ex1: If $a:5 = b:7 = c:8$ then $\frac{a+b+c}{a} = ?$

$a, b, c \rightarrow$ in terms of k

Ex2: If $p:q = r:s = t:u = 2:3$ then $(mp+nr+ot):(mq+ns+ou) = ?$

No need of k

since constant is given

Ex3: If $x:y=y:z=z:u = k$ and $x:u = 64:27$. The value of $x:z$ is ?

Ratio + Prop

Ex4: A sum of Rs 427 is to be divided among A, B and C in such a way that 3 times A's share, 4 times B's share and 7 times C's share are all equal. Find the share of each

① A, B, C $\rightarrow$ in terms of $k \rightarrow A:B:C$

② Thick { Ratio as part of total }

Ex5: $\frac{a-b}{a+b} = \frac{1}{5}$. Find $\frac{a^2-b^2}{a^2+b^2}$

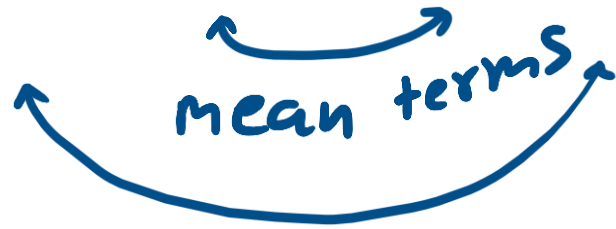
(m1) Concepts

(m2) Trick $\rightarrow$ {Substitution}

4] Theory Points in Proportion:

①

$$a : b :: c : d$$



Extremes

②

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

$\therefore$ Product of extremes =
Product of means

③ $\frac{a}{b} = \frac{c}{d} \rightarrow$ is fourth proportional

④ $\frac{a}{b} = \frac{b}{c} \rightarrow$ is third proportional {continued prop.}

⑤ mean proportional between a & b is $\sqrt{ab}$

* ⑥ Law of Equal Ratio :

$$1) \frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}$$

$$= \frac{a_1 + a_2 + \dots + a_n}{b_1 + b_2 + \dots + b_n}$$

$$= \frac{a_1 - a_2 - \dots - a_n}{b_1 - b_2 - \dots - b_n}$$

$$2) \frac{a}{b} = \frac{c}{d} \Rightarrow \frac{la}{lb} = \frac{mc}{md} = \frac{la \pm mc}{lb \pm md}$$

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5] Variation

① Direct proportion : $A \propto B$

$$\therefore A = k \cdot B$$

Ex : $\uparrow$ study , $\uparrow$ marks

$\uparrow$ Burgers , $\uparrow$ weight

NOTE: ① $A = k\beta$

$$\therefore \frac{A}{\beta} = k = \text{constant}$$

② * If 2 quantities of A & B then

$$\frac{A_1}{\beta_1} = \frac{A_2}{\beta_2}$$

② Inverse proportion : $A \propto \frac{1}{B}$

(Indirect
proportion)

$$\therefore A = k \cdot \frac{1}{B}$$

Ex : $\uparrow$ Gf, $\downarrow$ Study

$\uparrow$ Exercise, $\downarrow$ Disease

NOTE: ① $A = k/B$

$\therefore AB = k = \text{constant}$

② * If 2 quantities of A & B then

$$A_1 B_1 = A_2 B_2$$

③ Joint variation :

$$A \propto BC$$

$$\therefore A = kBC \quad \dots \quad \{k = \text{constant}\}$$

NOTE :

$$\textcircled{1} \quad A \propto B, B \propto C \rightarrow A \propto C$$

$$\textcircled{2} \quad A \propto B, A \propto C \rightarrow A \propto BC$$

$$\textcircled{3} \quad A \propto BC \rightarrow \frac{A}{C} \propto B \quad \textcircled{or} \quad \frac{A}{B} \propto C$$

6] EXAMPLES ON VARIATION:

Ex1: If m is proportional to n , $m=5$, $n=4$. Find m when $n=18$

{# By default : Direct proportional }

solⁿ:

① $m \propto n$

$\therefore m = kn$

② $m=5, n=4 \Rightarrow 5 = k \cdot 4 \Rightarrow k = 5/4$

③ $m = \frac{5}{4}n$ $\xrightarrow{n=18}$ $m = \frac{5}{4}(18) = 22.5 //$

Ex2: If x is inversely proportional to y and y is proportional to z , then xz

- a) Is proportional to y**
- b) Is inversely proportional to y**
- c) Is constant**
- d) Is proportional to y^2**

Ex2: If x is inversely proportional to y and y is proportional to z , then xz

- a) Is proportional to y
- b) Is inversely proportional to y
- ☒ c) Is constant
- d) Is proportional to y^2

Given : $x \propto 1/y \rightarrow x = k/y$

$y \propto z \rightarrow y = l z \rightarrow z = y/l$

$\therefore xz = \left(\frac{k}{y} \right) \left(\frac{y}{l} \right) = \frac{k}{l} = \text{constant}$

Ex3: The resistance of a wire is directly proportional to its length and inversely proportional to the square of its radius . Two wires of the same material have the same resistance and their radii are in the ratio of 9:8. If the length of the first wire is 162 cm, find the length of the other wire .

Solⁿ : ① $R \propto l$ and $R \propto \frac{1}{r^2}$

$$R_1 = R_2$$

$$r_1 : r_2 = 9 : 8 \rightarrow r_1 = 9x, r_2 = 8x$$

$$l_1 = 162, l_2 = ?$$

$$\textcircled{2} \quad \therefore R \propto \frac{l}{r^2} \rightarrow R = \frac{k l}{r^2}$$

$$R_1 = \frac{k l_1}{r_1^2}, \quad R_2 = \frac{k l_2}{r_2^2}$$

$$\therefore R_1 = R_2$$

$$\therefore \frac{\cancel{k} l_1}{r_1^2} = \frac{\cancel{k} l_2}{r_2^2}$$

$$\therefore \frac{l_1}{r_1^2} = \frac{l_2}{r_2^2}$$

$$\therefore \frac{162}{(9x)^2} = \frac{l_2}{(8x)^2}$$

$$\therefore l_2 = \frac{8^2 \times 162}{9^2} = 128 //$$

Ex4: If A varies directly to C and B also varies directly to C , which one of the following is not correct ?

a) $(A+B) \propto C$

b) $(A-B) \propto 1/C$

c) $\sqrt{AB} \propto C$

d) $A/B = \text{Constant}$

**Ex4: If A varies directly to C and B also varies directly to C ,
which one of the following is not correct ?**

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c) $\sqrt{AB} \propto C$

d) $A/B = \text{Constant}$

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Given : $A \propto C \rightarrow A = kC$

$$B \propto C \rightarrow B = lC$$

$$a) (A+B) = kC + lC = C(k+l)$$

$$\therefore (A+B) = C [\text{constant}] \Rightarrow A+B \propto C$$

$$\checkmark b) (A-B) = kC - lC = C(k-l)$$

$$\therefore (A-B) = C [\text{constant}] \Rightarrow (A-B) \propto C$$

$$c) \sqrt{AB} = \sqrt{kC \cdot \frac{1}{k}C} = \sqrt{C^2} = C$$

$$\therefore \sqrt{AB} = (\text{constant}) C$$

$$\therefore \sqrt{AB} \propto C$$

$$d) \frac{A}{B} = \frac{kC}{\frac{1}{k}C} = \frac{k}{\frac{1}{k}} = \text{constant}$$

7] Partnership

① Profit = Capital $\times$ Time

If A, B, C invested C_1, C_2, C_3 for time t_1, t_2, t_3 then their

profit share ratio = $C_1 t_1 : C_2 t_2 : C_3 t_3$

$(P_1 : P_2 : P_3)$

Ex1: A and B enters in a partnership with Rs 8000 and Rs 15000 respectively. After 3 months C joins them by investing Rs 10000. 4 months before the first year is completed, B quits. In what ratio should the profit of Rs 255000 earned in first year be distributed among the three? Also find their individual shares

Solⁿ:

Persons	Capital	Time (months)
A	8000	12
B	15000	12 - 4 = 8
C	10000	12 - 3 = 9

② Profit share of A, B, C $\Rightarrow$

$$P_A : P_B : P_C = C_A T_A : C_B T_B : C_C T_C$$

$$\begin{aligned}\therefore P_A : P_B : P_C &= 8000 (12) : 15,000 (8) : 10,000 (9) \\ &= 16 : 20 : 15\end{aligned}$$

③ Total Profit = 2,55,000

$$P_A : P_B : P_C = 16 : 20 : 15$$

① (Milk)

② (Std Approach)

$$\text{Total Profit} = 2,55,000$$

$$P_A + P_B + P_C = 2,55,000$$

$$16x + 20x + 15x = 2,55,000$$

$$x = 5000$$

$$A's \text{ share} = 16x = 16(5000) = 80K$$

$$B's \text{ share} = 20x = 20(5000) = 100K$$

$$C's \text{ share} = 15x = 15(5000) = 75K$$

Ex2: A person P started a business with a capital of Rs 2525 and another person Q joined P after some months with a capital of Rs 1200. Out of the total annual profit of Rs 1644, P's share was Rs 1212. When did Q join as partner?

Solⁿ : ①

Person	Capital	Time (months)
P	2525	12
Q	1200	12 - x

$$\begin{aligned}
 \textcircled{2} \quad P : Q &= C_P T_P : C_Q T_Q \\
 &= \frac{2525(12)}{300} : \frac{1200(12-x)}{300} \\
 &= 101 : 4(12-x)
 \end{aligned}$$

③ $P_P : P_Q = 101 : 4(12-x)$

Total profit = 1644

P's profit = 1212

$\therefore \frac{101}{101 + 4(12-x)} \times \frac{1644}{137} = \frac{1212}{1}$

$\therefore 137 = 101 + 4(12-x)$

$\therefore x = 12/4 = 3 //$

8] Homework Questions:

Ex1:

Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?

5 : 7 : 8

20 : 49 : 64

38 : 28 : 21

None of these

Ex2:

If $2A=3B=4C$, then $A:B:C$ is

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Ex3:

If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, **then** $\frac{b^3+c^3+d^3}{a^3+b^3+c^3} = ?$

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Ex4:

If $\frac{x}{2} = \frac{y}{3} = \frac{z}{4} = \frac{2x-3y+5z}{k}$ **then k = ?**

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Ex5:

The ratio between the third proportional of 12 and 30 and mean proportional of 9 and 25 is

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9] Homework Questions Solution:

Ex1:

Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?

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None of these

Ex1:

Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?

5 : 7 : 8

20 : 49 : 64

38 : 28 : 21

None of these

$$\textcircled{1} \quad P_1 : P_2 : P_3 = 5 : 7 : 8$$

$$T_1 : T_2 : T_3 = 14 : 8 : 7$$

$$\textcircled{2} \quad P_1 : P_2 : P_3 : C_1 T_1 : C_2 T_2 : C_3 T_3$$

$$5 : 7 : 8 = C_1 (14) : C_2 (8) : C_3 (7)$$

$$5 : 7 : 8 = C_1(14) : C_2(8) : C_3(7)$$

Convert C_2 & $C_3 \rightarrow$ in terms of C_1

$$\frac{5}{7} = \frac{14C_1}{8C_2} \Rightarrow C_2 = \frac{7 \times 14C_1}{8 \times 5} = \frac{49}{20} C_1$$

$$\frac{5}{8} = \frac{14C_1}{7C_3} \Rightarrow C_3 = \frac{8 \times 14C_1}{7 \times 5} = \frac{16C_1}{5}$$

$$\therefore C_1 : C_2 : C_3 = C_1 : \frac{49}{20} C_1 : \frac{16}{5} C_1$$

$$= 1 : \frac{49}{20} : \frac{16}{5}$$

$$= 20 : 49 : 64$$

Ex2:

If $2A=3B=4C$, then $A:B:C$ is

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Ex2:

If $2A=3B=4C$, then $A:B:C$ is

$$\textcircled{1} \quad 2A = 3B = 4C = K$$

$$A = K/2, \quad B = K/3, \quad C = K/4$$

$$\therefore A : B : C = \frac{K}{2} : \frac{K}{3} : \frac{K}{4}$$

$$= \frac{12K}{2} : \frac{12K}{3} : \frac{12K}{4}$$

$$= 6 : 4 : 3 //$$

Ex3:

If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, **then** $\frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3} = ?$

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Ex3:

If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, then $\frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3} = ?$

$$\textcircled{1} \quad \frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k$$

$$a = bk, \quad b = ck, \quad c = dk$$

$$\textcircled{2} \quad \frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3} = \frac{b^3 + c^3 + d^3}{(bk)^3 + (ck)^3 + (dk)^3}$$

$$\textcircled{\checkmark} \quad \frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3} = \frac{b^3 + c^3 + d^3}{(bk)^3 + (ck)^3 + (dk)^3}$$

$$= \frac{1}{k^3}$$

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$$\textcircled{3} \quad \frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k$$

$$\therefore \frac{a}{b} \times \frac{b}{c} \times \frac{c}{d} = k \cdot k \cdot k$$

$$\therefore \frac{a}{d} = k^3$$

$$\therefore \frac{1}{k^3} = \frac{d}{a}$$

$$Ans = \frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3} = \frac{d}{a}$$

Ex4:

If $\frac{x}{2} = \frac{y}{3} = \frac{z}{4} = \frac{2x-3y+5z}{k}$ **then k = ?**

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Ex4:

If $\frac{x}{2} = \frac{y}{3} = \frac{z}{4} = \frac{2x-3y+5z}{k}$ then $k = ?$

$$\textcircled{1} \quad \frac{x}{2} = \frac{y}{3} = \frac{z}{4} = \frac{2x-3y+5z}{k} = 1$$

$$\therefore x = 2l, y = 3l, z = 4l$$

$$\therefore \frac{2x-3y+5z}{k} = \frac{2(2l)-3(3l)+5(4l)}{k}$$

$$= \frac{l}{k} (4-9+20) = \frac{15l}{k}$$

$$\therefore \frac{2x - 3y + 5z}{k} = 1$$

$$\therefore \frac{15l}{k} = 1$$

$$\therefore k = 15$$

Ex5:

The ratio between the third proportional of 12 and 30 and mean proportional of 9 and 25 is

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Ex5:

The ratio between the third proportional of 12 and 30 and mean proportional of 9 and 25 is

① let third prop. be x

$$\therefore \frac{12}{30} = \frac{30}{x} \rightarrow x = 75$$

② mean prop = $\sqrt{9 \times 25} = 15$

③ Ratio = $\frac{75}{15} = \frac{5}{1}$ //